

Applied Optimization

Exercise 8

Trust Region and Equality Constrained Optimization

What to hand in

A .zip compressed file renamed to `Exercisen-Group.zip` where n is the number of the current exercise sheet. It should contain:

- Hand in **only** the files you changed (headers and source). It is up to you to make sure that all files that you have changed are in the zip.
- A `readme.txt` file containing a description on how you solved each exercise (use the same numbers and titles) and the encountered problems. Indicate what fraction of the total workload each project member contributed.
Failure to do so will result in you losing a point.
- Other files that are required by your `readme.txt` file. For example, if you mention some screenshot images in `readme.txt`, these images need to be submitted too.
- For the theory exercise, put `TheoryExercise.pdf` with your solutions in the same .zip you submit for the code.
- Submit your solutions to ILIAS before the submission deadline. Late submissions will receive 0 points!
- **Important:** Your submission MUST compile without any issue in order to obtain a passing grade.

Trust Region Subproblem: Exact Solution (1 + Bonus 1 pts)

Consider the following trust region subproblem in two dimensions, $v \in \mathbb{R}^2$:

$$\begin{aligned} & \text{minimize } m(v) = g^T v + \frac{1}{2} v^T B v \\ & \text{subject to: } \|v\| \leq \Delta \end{aligned}$$

as discussed in [Nocedal, Sec 4.3] and the slides. In every problem, the inputs B , g and Δ are given. Plot the quadratic model and the trust region with the **tool**. Compute the λ^* and v^* which satisfy the KKT optimality conditions.

Solution:

For problems 3 to 7 we summarize here the main ideas of the exact solution classification. (Theorem 4.1, pag. 70. Nocedal)

v^* is a global solution to the trust region problem above if and only if v^* is feasible and there exist a $\lambda \geq 0$ for which the following conditions are satisfied.

1. $(B + \lambda I)v^* = -g$
2. $\lambda(\Delta - \|v^*\|) = 0$
3. $(B + \lambda I)$ is positive definite

So the calculations should follow the following case distinctions:

case 1: if $\|v^*\| \leq \Delta$ then $\lambda = 0$ and solution is unique if B is positive definite.

case 2: if $\|v^*\| > \Delta$, λ^* is a root of the function:

$$\|v(\lambda)\|^2 - \Delta^2 = 0$$

with

$$\|v(\lambda)\|^2 = \sum_{j=1}^n \frac{(q_j^T g)^2}{(\lambda_j + \lambda)^2}$$

For sorted eigenvalues $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$ and corresponding eigenvectors $q_1, q_2, \dots, q_n$. Given that the dot product of the eigenvector corresponding to the smallest eigenvalue with g is not zero $q_1^T g \neq 0$.

If $q_1^T g = 0$, $(B - \lambda_1 I)$ is singular, so there exist a vector z such that $\|z\| = 1$ and $(B - \lambda_1)z = 0$, z is an eigenvector of B and is orthogonal to the other q_j for $j \neq 1$.

$$v(\lambda) = \sum_{j:\lambda_j \neq \lambda_1} \frac{-q_j^T g}{\lambda_j + \lambda} q_j + \tau z$$

for any scalar τ , and

$$\|v(\lambda)\|^2 = \sum_{j:\lambda_j \neq \lambda_1} \frac{(q_j^T g)^2}{(\lambda_j + \lambda)^2} + \tau^2$$

In the following we identify in which case each of the problems, sketch and compute the solutions from the formulas above.

Problem 1:

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \Delta = 1/2$$

Compute an approximation of v^* with Cauchy point method.

Solution:

First of all, we check

$$g^T B g = (11) \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 6 > 0$$

We need to analytically find the minimum $t > 0$ of $\phi(t) = m(-t \frac{g}{\|g\|})$ and truncate w.r.t. Δ . Thus,

$$v^C = -t\Delta \frac{g}{\|g\|}$$

, with $t = \min\{\frac{\|g\|^3}{\Delta g^T B g}, 1\} = \frac{2\sqrt{2}}{3} \approx 0.9428$.

$$\text{So } v^C = -\frac{2\sqrt{2}}{3} * \frac{1}{2} * \frac{\sqrt{2}}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1/3 \\ -1/3 \end{pmatrix}$$

Problem 2:

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \Delta = 1/2$$

Compute an approximation of v^* with the dogleg method.

Solution:

First compute v^N and v^G ,

$$B^{-1} = \begin{pmatrix} 1/4 & 0 \\ 0 & 1/2 \end{pmatrix}$$

$$v^N = -B^{-1}g = \begin{pmatrix} -1/4 \\ -1/2 \end{pmatrix}, \quad v^G = -\frac{g^T g}{g^T B g} g = -\frac{2}{6}g = \begin{pmatrix} -1/3 \\ -1/3 \end{pmatrix}$$

Since $\|v^N\| = 0.559 > 1/2$ and $\|v^G\| = 0.4714 < 1/2$, the second segment intersect with the trust region boundary. So $v^D = v^G + (\tau - 1)(v^N - v^G)$, with τ satisfying

$$\|v^G + (\tau - 1)(v^N - v^G)\| = 1/2$$

Write out the equation, it is

$$(\tau/6 + 1/6)^2 + (\tau/12 - 5/12)^2 - 1/4 = 0$$

Solving the above equation for $1 \leq \tau \leq 2$, we get $\tau = 1.4$.

The v^D with Dogleg method is

$$v^D = v^G + (1.4 - 1)(v^N - v^G) = \begin{pmatrix} -0.3 \\ -0.4 \end{pmatrix}$$

Problem 3:

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \Delta = 1$$

Compute the exact solution.

Solution:

Start by checking whether condition 2 holds with $\lambda = 0$, that is when $\|v^*\| \leq \Delta$

$$B^{-1} = \begin{pmatrix} 1/4 & 0 \\ 0 & 1/2 \end{pmatrix}$$

$$B^{-1}g = \begin{pmatrix} 1/4 & 0 \\ 0 & 1/2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/4 \\ 1/2 \end{pmatrix}$$

$$\|B^{-1}g\| = \sqrt{1/16 + 1/4} = \frac{\sqrt{5}}{4} \approx 0.559... < \Delta = 1$$

Thus $\lambda = 0$ and $v^* = -B^{-1}g = (-1/4, -1/2)^T$. Figure 1 depicts the model function and the trust region centered without loss of generality at $(0,0)$. The model function is computed by plugging in the values of g and B and $v = (x,y)^T$.

$$\begin{aligned} m(v) &= (1,1) \begin{pmatrix} x \\ y \end{pmatrix} + \frac{1}{2}(x,y) \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \\ &= 2x^2 + y^2 + x + y \end{aligned}$$

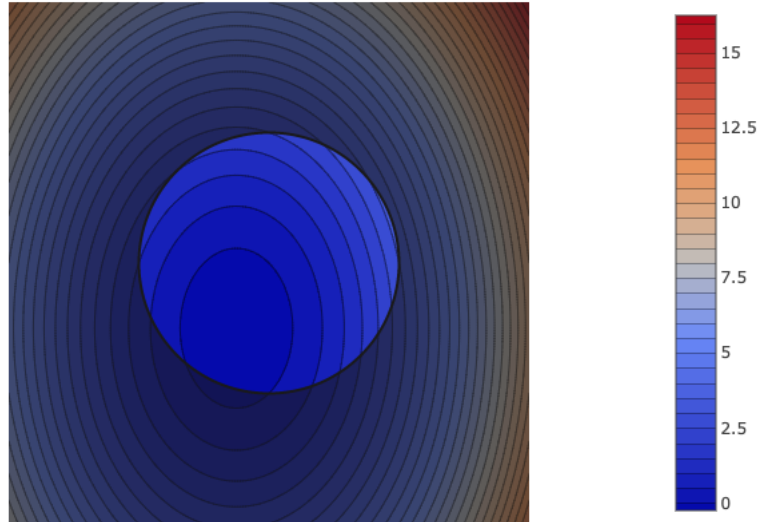


Figure 1: Problem 3: model $m(v)$ with trust region $\Delta \leq 1$ that includes the minimum of the convex quadratic model.

This function represents a paraboloid opening in the positive z direction.

Problem 4:

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \Delta = 1/2$$

Compute the exact solution.

Solution:

Following the slides, $\lambda_1 = 2$, $q_1 = (0, 1)^T$ and $\lambda_2 = 4$ and $q_2 = (1, 0)^T$. If we take v^* from problem 3, it is outside the trust region since $\|v^*\| = \frac{\sqrt{5}}{4} > \Delta = \frac{1}{2}$.

Thus we are in the case 2 of the classification and we need to solve for the zeros of $\|v(\lambda)\|^2 - \Delta^2 = 0$.

$$v(\lambda) = -\left(\frac{q_2^T g}{\lambda_2 + \lambda} q_2 + \frac{q_1^T g}{\lambda_1 + \lambda} q_1\right) = -\frac{1}{4 + \lambda} q_2 + -\frac{1}{2 + \lambda} q_1$$

$$\|v(\lambda)\|^2 - \Delta^2 = \frac{1}{(4 + \lambda)^2} + \frac{1}{(2 + \lambda)^2} - \left(\frac{1}{2}\right)^2 = 0$$

Simplify the above equation, we get

$$\lambda^4 + 12\lambda^3 + 44\lambda^2 + 48\lambda - 16 = 0$$

We can use a mathematical tool, e.g. matlab, to calculate the roots of the polynomial. Since $\lambda > -\lambda_1 = -2$, we take $\lambda = 0.2645$.

$$\text{Putting } \lambda = 0.2645 \text{ back to } v(\lambda), v^* = -\frac{1}{2+0.2645} \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \frac{1}{4+0.2645} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = -\begin{pmatrix} 0.2345 \\ 0.4416 \end{pmatrix}$$

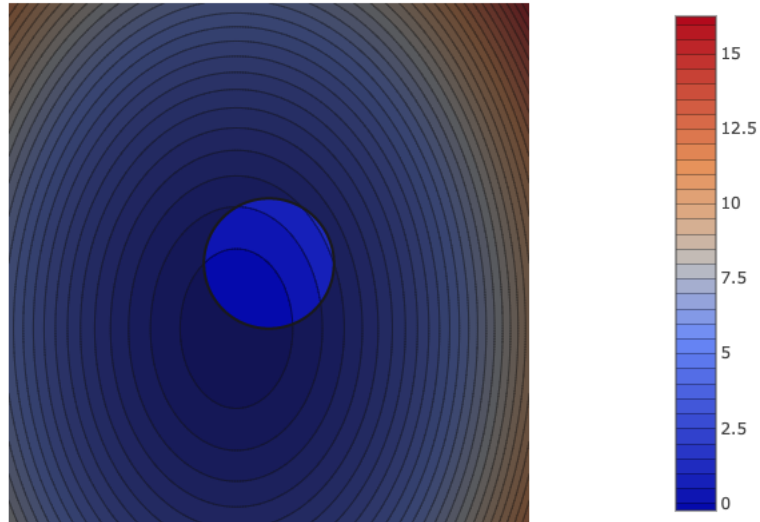


Figure 2: Problem 4: model $m(v)$ with trust region $\Delta \leq 1/2$ that excludes the minimum of the convex quadratic model.

Problem 5 (bonus):

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad \Delta = 1/2$$

Compute the λ^* and v^* which satisfy the KKT optimality conditions.

Solution:

$\lambda_1 = 2, q_1 = (0, 2)^T$ and $\lambda_2 = 4$ and $q_2 = (1, 0)^T$.

$$\|B^{-1}g\| = \sqrt{0 + 1/4} = 1/2 = \Delta$$

Thus we are in the case 1, resulting in

$$v(\lambda = 0) = \begin{pmatrix} -1/2 \\ 0 \end{pmatrix}.$$

Problem 6: (bonus)

Consider now the the indefinite matrix

$$\tilde{B} = \begin{pmatrix} 4 & 0 \\ 0 & -2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \Delta = 1$$

Compute the λ^* and v^* which satisfy the KKT optimality conditions.

Solution:

In this case $\|v^*\| \leq \Delta$ holds, but $(B + \lambda I)$ is not positive semi-definite (psd) with $\lambda = 0$. For it to be psd we need to solve for $\lambda \geq 2$. Since g is orthogonal to q_1

$$v(\lambda) = \frac{-q_2^T g}{\lambda_2 + \lambda} q_2 + \tau q_1 = \frac{-1}{4 + \lambda} q_2 + \tau q_1 = \begin{pmatrix} \frac{-1}{4 + \lambda} \\ \tau \end{pmatrix}.$$

With the difference that now the indefinite matrix B leads to the subregion problem not having a unique solution anymore.

The model equation can be obtained as before with

$$\begin{aligned} m(v) &= (1, 0) \begin{pmatrix} x \\ y \end{pmatrix} + \frac{1}{2} (x, y) \begin{pmatrix} 4 & 0 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \\ &= 2x^2 - y^2 + x \end{aligned}$$

The plot is as shown in Figure 3.

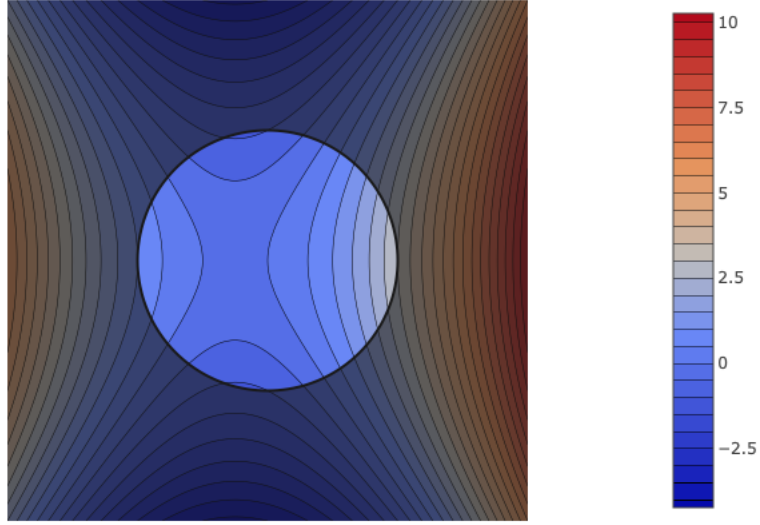


Figure 3: Problem 6: model $m(v)$ with trust region $\Delta \leq 1$ that includes the saddle of the quadratic model.

We can see that the model $m(v)$ decreases in the directions of $\pm q_1$, now we expect a non-zero value for τ^2 that would give us the two symmetric solutions on the boundary of the trust region. Thus

$$\begin{aligned} \|v(\lambda)\|^2 - \Delta^2 &= \frac{-1^2}{(4+\lambda)^2} + \tau^2 - 1 = 0 \\ \tau^2 &= 1 - \frac{1}{(4+\lambda)^2} \end{aligned}$$

Replace x, y with λ, τ , we get

$$m(v) = \frac{2}{(4+\lambda)^2} - \frac{1}{(4+\lambda)} - \tau^2 = \frac{3}{(4+\lambda)^2} - \frac{1}{(4+\lambda)} - 1.$$

The

$$m' = \frac{(4+\lambda) - 6}{(4+\lambda)^3}$$

Remember the conditions for λ and τ : $\lambda \geq 2$ and $0 < \tau^2 < 1$. We can get $m' \geq 0$ for $\lambda \geq 2$. We take the value of $\lambda = 2$ that make the function smallest. This is interpreted as $v(\lambda)$ laying on the boundary of the trust region:

$$\tau(\lambda = 2) = 1 - \frac{1}{36} = \frac{35}{36}$$

Thus, we get two solutions with $\lambda^* = 2$ and $\tau = \frac{\sqrt{35}}{6}, -\frac{\sqrt{35}}{6}$, $v^* = \begin{pmatrix} -1/6 \\ \pm\sqrt{35}/6 \end{pmatrix}$.

Problem 7: (bonus)

Same parameters as in problem 6, but now the gradient is zero $g = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Solution:

In this case, conditions 1 and 2 could be satisfied with $\lambda = 0$ because $\|B^{-1}g\| = 0 < \Delta = 1$. But $(B + \lambda I)$ is not psd with $\lambda = 0$, it imposes the condition $\lambda \geq 2$. Thus we need to solve the zeros of $\|v(\lambda)\|^2 - \Delta^2 = 0$ for $\lambda \geq 2$. Additionally, since $q_1^T g = 0$ and $q_2^T g = 0$, we need to add two arbitrary scalars τ_1 and τ_2 and solve for the trust region problem to find the minima of the model $m(v)$. The model is in this case the same saddle as before but centered at $(0,0)$ as depicted in Figure 4. With equation:

$$m(v) = 2x^2 - y^2$$

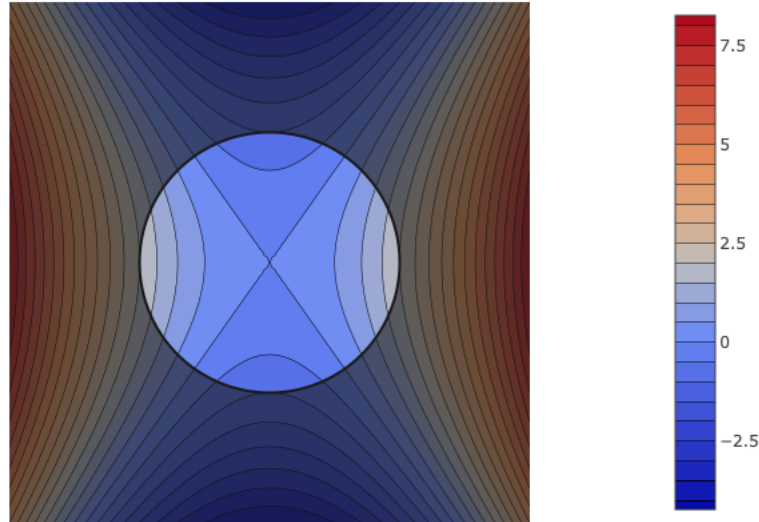


Figure 4: Problem 7: model $m(v)$ with trust region $\Delta \leq 1$ centered at the saddle point of the quadratic model.

$$v(\tau_1, \tau_2) = \tau_1 q_1 + \tau_2 q_2$$

and

$$\|v(\lambda)\|^2 - \Delta^2 = \tau_1^2 + \tau_2^2 - 1 = 0$$

As we know from our interpretation of the eigenvalue decomposition of B , the minima of $m(v)$ are located at $v^* = \pm q_1$. To verify this consider the equation resulting from the first condition:

$$\begin{aligned}
(B + \lambda)v^* &= -g \\
(B + \lambda)(\tau_1 q_1 + \tau_2 q_2) &= \vec{0} \\
\tau_1(\lambda_1 + \lambda)q_1 + \tau_2(\lambda_2 + \lambda)q_2 &= \vec{0}
\end{aligned}$$

Taking the euclidean norms on both sides we obtain the scalar equation:

$$\tau_1^2(\lambda + \lambda_1)^2 + \tau_2^2(\lambda + \lambda_2)^2 = 0$$

Defining $\alpha := \tau_1^2$ and $\beta := \tau_2^2$ the trust region constraint can be rewritten as $\alpha + \beta = 1$ with α and $\beta \in [0, 1]$, thus

$$\alpha(\lambda - 2)^2 + (1 - \alpha)(\lambda + 4)^2 = 0$$

which simplifies to

$$\lambda^2 + (8 - 12\alpha)\lambda + (16 - 12\alpha) = 0,$$

and whose roots are

$$\lambda^* = 6\alpha - 4 \pm 6\sqrt{(\alpha - 1)\alpha}.$$

For λ to be real, $\alpha = 1$, thus $\lambda^* = 6 - 4 = 2$ which satisfies the trust region condition with equality, and $\beta = 0$. $\tau_1 = \pm 1$, $\tau_2 = 0$ and $v^* = \begin{pmatrix} \pm 1 \\ 0 \end{pmatrix}$.